

MUHAMMAD MUGHEES AHMED

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Objective

AI/ML Engineer with hands-on experience in **Python**, **Django**, and **FastAPI**, specializing in building scalable backend systems and intelligent applications. Skilled in **machine learning**, **LLM integration**, and **data processing pipelines** using tools like **LangChain** and **LangGraph**. Strong foundation in **NLP techniques** including **TF-IDF**, **Cosine Similarity**, and **XGBoost** for real-world problem solving. Experienced in designing end-to-end AI workflows, integrating APIs, and optimizing model performance through **prompt engineering** and evaluation techniques. Passionate about developing production-ready AI solutions and leveraging data-driven approaches to solve complex challenges.

Education

Bachelor of Computer Science | CGPA: 3.37

SEP 2021 – JUNE 2025

Bahria University Lahore , Pakistan

Relevant CourseWork: Artificial Intelligence, Data Science, Data Structures & Algorithms, Database Management Systems, Programming Fundamentals, Object-Oriented Programming, Operating Systems, Big Data Analytics, Web Engineering, Software Testing

Professional Experience

AI Engineer | Techtics.ai

JAN 2026 - Present

- Developed and deployed **AI-driven backend systems** using **Python** and **Django**, focusing on scalability, maintainability, and efficient handling of real-world data workflows.
- Built and optimized **web scraping and data pipelines** to extract, clean, and process large volumes of structured and unstructured data for downstream AI applications.
- Designed and implemented **LLM-based solutions** using **LangChain** and **LangGraph**, applying advanced **prompt engineering** and integrating APIs to deliver reliable, production-level AI features.

Python Developer Intern | TRIGENT Inc.

AUG 2024 – NOV 2024

- Engineered robust backend systems using **FastAPI**, **SQLAlchemy**, and **SQLite**, enabling efficient API development and seamless data management for scalable applications.
- Developed and optimized **data processing workflows** and backend logic to support real-time functionality and improve system performance.
- Contributed to **LLM training pipelines** by working on **RLHF** and **Supervised Fine-Tuning (SFT)**, enhancing model alignment through **prompt evaluation**, ranking, and preference modeling techniques.

Skills

Programming Languages: Python, Java, C/C++, JavaScript

Frameworks: DjangoRest, Django, FastAPI, nodeJS

Data Analysis & Machine Learning: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TF-IDF, XGBoost

LLM & AI Tools: LangChain, LangGraph, Prompt Engineering, NLP, Model Evaluation

Database: Postgres, ChromaDB, MongoDB, MySQL, PHPmyadmin

Tools & Platforms: Git, Github, VS Code, Linux, Postman, Google Colab

Certifications

- AI Developer Professional Course - **IBM**
- Python for AI, Data Science, and Development - **IBM**
- Building Generative AI-Powered Applications with Python - **IBM**
- ChatGPT Prompt Engineering – **deepLearning.ai**
- LangChain & LangGraph – **deepLearning.ai**
- Data Visualization – **Kaggle**
- Introduction to Machine Learning – **Google Developers**
- Machine Learning Crash Course – **Google Developers**

Projects

Final Year Project: Automated HR System

Tech Stack: ReactJS, NodeJS, ExpressJS, MongoDB, Python, FastAPI, XGBoost, TF-IDF, Cosine Similarity

- Led development of a **full-stack MERN application** integrating end-to-end AI/ML workflows for HR automation.
- Implemented an **AI-based Resume Shortlisting & Ranking System** using **XGBoost** and **TF-IDF/Cosine Similarity** for candidate relevance scoring.
- Developed a scalable backend using **FastAPI** to serve **RESTful APIs** for real-time data processing and seamless integration with the MongoDB database.
- Designed and deployed an **AI-driven HR Chatbot** using Python (MERN + Python stack) for automated employee onboarding and query resolution.

Q&A RAG System for Document Search

Tech Stack: Python, LangChain, FastAPI, ChromaDB, Streamlit

- Developed a basic **Retrieval-Augmented Generation (RAG) system** in Python to enable contextual Q&A over custom documents.
- Utilized **LangChain** to manage the end-to-end pipeline, including document loading, chunking, embedding, and retrieval from a vector database.
- Exposed the RAG core logic via a high-performance **FastAPI** endpoint for real-time query processing.
- Created a simple web interface (Streamlit) for users to upload documents and interact with the Q&A chatbot.

C++ Projects

1. Rock Scissors Paper Game (C++)

- Built an interactive **CLI game** using control structures and **randomization** in C++.

2. Stock Management System (C++)

- Developed a **trade tracking system** using **OOP** and file handling in C++.

3. Cafe Management System (C++)

- Implemented **doubly linked list** logic for menu, billing, and queue simulation.

4. Mini Compiler (C/C++)

- Created a **lexical analyzer** using **regex** and **DFAs** to tokenize source code.

5. Rental System using Banker's Algorithm (C)

- Simulated **OS-level resource allocation** to avoid deadlock conditions.

Extracurricular

- RIZQ BULC | 2022 – 2024
- AL-KHIDMAT Foundation | 2024 – Present
- Vice Head of Table Tennis | BULC | 2023 - 2024